

DEPARTMENT OF MINING ENGINEERING

Mine Ventilation Practical-I

(MNC 210)

Mine Ventilation Laboratory



I.I.T (ISM), DHANBAD

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Experiment No-01

Aim : - Constructional features and applications of Flame Safety Lamp.

Theory:-

Historical development

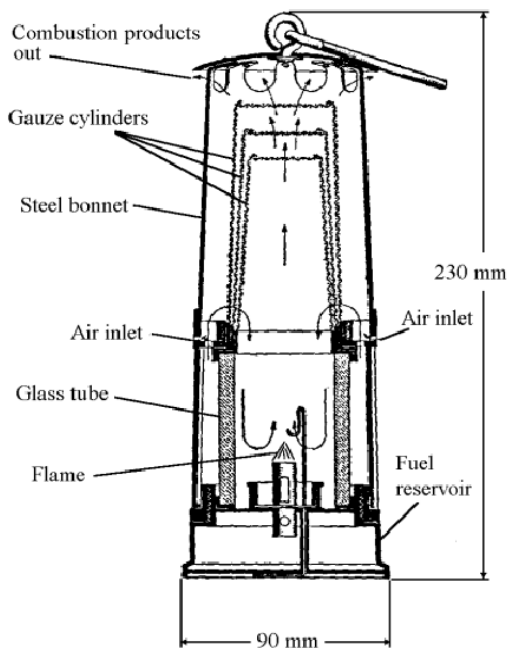
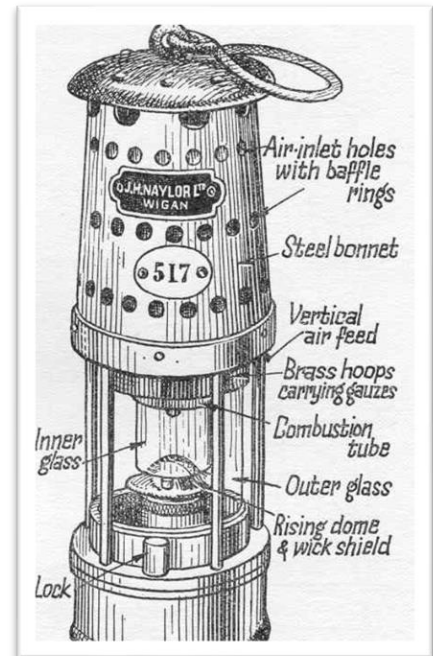
To get rid of the recurring problem of firedamp explosion due to open lamp, Sir Humphry Davy, in 1815, came forward with a revolutionary device known as Davy Safety Lamp. The flame of the lamp was covered by a copper wire gauze having good thermal conductivity. The flame heat was dissipated throughout the wire gauze. Thus not enough heat passed to ignite the explosive mixture surrounding the lamp. The light output of Davy lamp was poor as a fine wire gauze surrounded the whole flame. In mid nineteenth century, Dr. Clanny modified the design by surrounding the flame with a cylindrical glass and putting the wire gauze over the glass. There was tremendous increase in light. But the lamp was still not safe in a strong current of explosive firedamp mixture. Marsaut improved upon the design by the late nineteenth century by providing a bonnet surrounding the wire gauze. He strengthened the safety by introducing more than one wire gauze. Marsaut lamp forms the basis of modern design of flame safety lamp.

Modern Flame Safety lamp consists of

Marsaut principal with the incorporation of re-lightening devices. Safety locks to prevent unauthorized openings, controlled air feedings arrangement from top, middle and bottom part of this flame lamp with the introduction of general illumination by electric lamps & with the development of electric cap lamp the flame safety lamp has been suppressed as a source of illumination in mines. Presently, these are manufactured by JK Dey & Sons, Kolkata & marketed in the trade name Velox as approved by DGMS.

Classification of flame safety lamps

- I. Based on the provision of combustion chimney
 - (i) Safety lamp with combustion chimney
 - (ii) Safety lamp without combustion chimney.
- II. Based on Internal Relighting Arrangement
 - (i) Lamp with Internal relighting arrangement.
 - (ii) Lamp without Internal relighting arrangement.
- III. Based on the position of entry of air
 - (i) Bottom- Feed type.
 - (ii) Mid- Feed type.
 - (iii) Selective top / bottom- feed type



Different components of Flame Safety Lamp:-

Carrying hook, Vent holes, Bonnet, Deflector ring, Waffel ring containing inlet holes, Pillars (5 Nos.), Magnetic lock, Outer gauze, Inner gauze, Combustion Chimney, Inner glass, Asbestos rings, Outer Glass, Glass retainer, Wick retainer, Wick holder, Non-spill type oil valve, Wick, Wick adjuster, Oil vessel.

Construction:-

A Flame Safety Lamp consists of 3 main sections which are tightly screwed together:-

Lower Section:-

This section contains the oil vessel, burner, and wick assembly. A pointer passes through a hollow cylinder welded in the oil vessel. This pointer can regulate the flame height without opening the lamp. A spring loaded pointer moves from one groove to another of ratchet while fitting. No internal re-lightening arrangement is there in GI-5. Wick may either be flat or round type, and oil used in oil vessels are:-

White kerosene, Esso Solvent Sprit No. 1425, solvent SBP 55/100, Petrol, etc.

Middle Section:-

This consists of 2 rings, the lower and the middle ring interconnected by the steel rods used to protect the glass placed around flame. The lower ring is provided with air inlet holes unless the lamp is of top feed type. When the air enters through the parts in the upper section. In most gas testing lamps, only one cylinder surrounds the flame. However, better illumination is provided if an inner glass connected to a chimney above provides a separate path for the fresh air & without mixing with the products of combustion. The outer glass is supplied with a thin asbestos ring both at top and bottom edge to make it airtight.

Upper Section:

This section comprises a bonnet and protects the wire gauge enclosing two steel wire gauzes, usually at 28 mesh size, which is placed over the glass cylinder with an

intervening asbestos washer & a hinge handle is also provided for holding the lamp. Some models, e.g., the Velox GL-5 combustion chimney, are supplied inside the wire gauze. At the top of the combustion chimney, there is a cover plate. This canopy prevents the hot air from releasing directly at the top of the wire gauze. The canopy retains the unburnt carbon particles and protects the wire gauze from carbon deposits.

Safety Features:-

Flame Safety lamp is provided with several significant safety features such as:-

- (a) Presence of a wire gauzes in the upper section of the lamp to increase the surface area for heat dissipation and absorb the heat to prevent explosion.
- (b) There is a presence of 5 rods in the middle section of the lamp used to protect the glass from unwanted stretch by the straight edge.
- (c) Presence of asbestos ring at the above below portion of the outer glass. So as to provide airtightness, thus the chances of the explosion are reduced.
- (d) Presence of magnetic locking in the lower section of the lamp to prevent the unauthorized opening of the Flame Safety lamp immediately.

Purpose of Flame Safety Lamp:-

The purpose of the Flame Safety lamp is now:

- Detect the oxygen deficiency in mine.
- Detect the presence of CO₂.
- Detect and estimate the percentage of methane in the mine.
- Provide illumination.

Experiment No:-02

Aim: - Gas testing with flame safety lamp.

Apparatus:-

- (i) Flame Safety lamp.



- (ii) Gas testing chamber.

Theory: -

Two types of Flames used For Safety lamps: Luminous or normal flame and Non-luminous or standard or testing flame.

Normal Flame:-

A lamp is carried in mine with a normal flame under the condition, the lamp burns with an elongated flame without producing any soot. It consists of 3 zones:

1. The inner zone of non-combustion
2. Intermediate zone of incomplete combustion
3. The outer zone of complete combustion

Standard Flame:-

It is obtained by lowering the wick until all white light disappears, leaving a well-defined blue line across the top of the flame just above the speck of yellowish light.

Fuel Cap & Gas Cap: - While testing firedamp with a flame safety lamp, it is of primary importance to differentiate between fuel cap & gas cap. The fuel cap is produced by burning the fuel alone in an atmosphere not containing flammable gas. When the lamp burns in the atmosphere containing flammable gas, it gives a pale blue cap over the Fuel cap called the gas cap.

The height and shape of the gas cap depend upon the following factors:

- (a) **The shape of Wick:** - Wick may be either flat or round. Preparation of round wick is easier, but Flat wick is claimed to be a superior indicator of Firedamp concentration.
- (b) **Type of fuel:** - The fuel cap and the gas cap height are more with more volatile fuel than a less volatile one.
- (c) **Width of the Wick:** - Wider the wick, longer is the cap.

There are two types of test that can be performed for detecting and estimating the presence of firedamp:

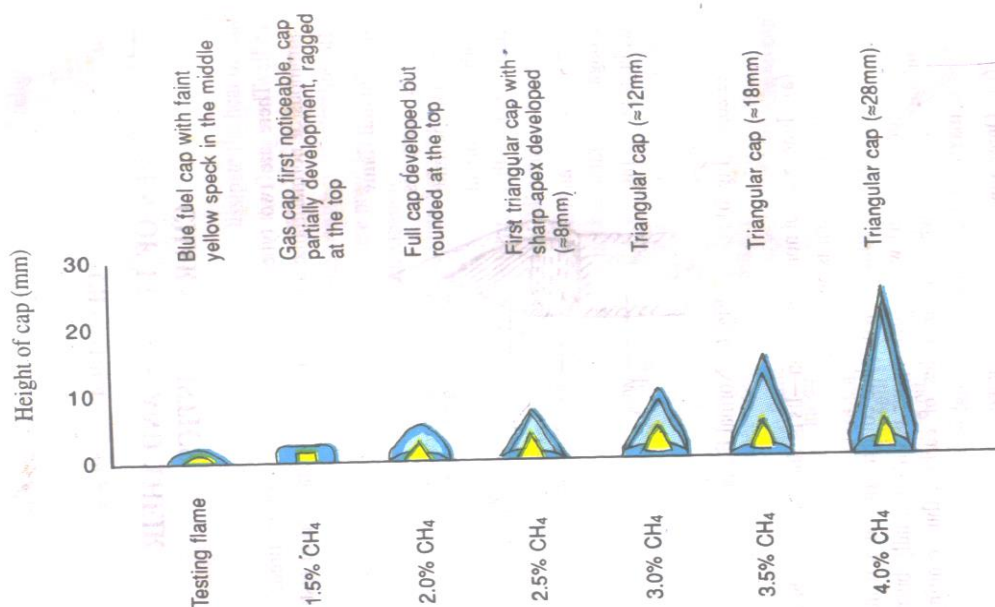
Accumulation Test:-

In the unventilated or comparatively less ventilated areas such as blind Heading, goaf edged, and the face of a rising gallery, the gas concentration varies wide along the vertical plane. The topmost air layer may contain almost pure methane. In normal flame, when methane burns inside the lamp, it is filled with products of combustion which helps in extinguishing the flame in higher methane concentration. In this test, the flames give a positive response indicating an accumulation of gas not less than 3%.

Percentage test:-

When an accumulation of a firedamp is suspected, it is always to go for the accumulation test followed by the percentage test only when the former gives a negative response.

For in the Event of a negative response to the accumulation test, the atmosphere still can contain fire damp below 3% level. A standard flame is required for the percentage test. The height and shape of the gas cap over the fuel cap measure methane, gas can hardly be identified. Below 1.5% methane, the flame can hardly be identified. Above methane concentration of 4%, the flame tip is generally not visible as it becomes longer than the glass height.



Test procedure in the laboratory

1. A flame safety lamp is put inside a test chamber connected with a gas cylinder through regulatory valves. The front side of the chamber is made of transparent plastic.
2. The surrounding is made dark before carrying out this test.
3. This test is carried out with a reduced flame. The flame can be reduced with the help of the regulating knob.
4. The flame should be reduced to an extent such that there appears a continuous blue line just above a speck (a very small mark or shape) of white/yellow light.
5. The percentage of methane is indicated by the height of the cap produced. The height of the cap produced is dependent on the following factors:
 - Width of the wick: The wider the wick, the longer is the cap.
 - Nature of fuel: Lesser is the boiling point of the fuel used, the longer is the cap produced.
 - Shape of the wick: Flat wick gives more accurate results compared to round ones.

Test procedure in the field

For performing an accumulation test in a suspected place of the mine gallery, the lamp with normal flame is gradually raised towards the roof and the change in the flame, if any, is observed. As soon as the flame starts 'spiring' or 'jumping' indicating a concentration of not less than 3% methane, the lamp is slowly brought down. No attempt should be made to raise the lamp any further, otherwise the lamp would most likely extinguish at higher concentration.

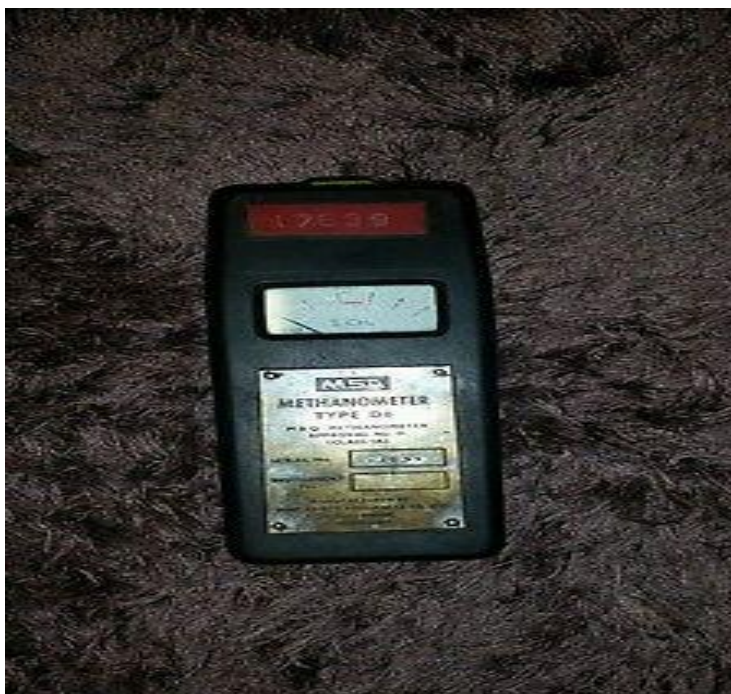
Limitation of flame safety lamp:-

It can't measure methane concentration below 1.5%.

Experiment No:-03

Aim: -Detection of CH₄ in mine atmosphere using Methanometer.

Apparatus used: - MSA (Mine Safety Appliance) D6 Methanometer.



MSA D6 Methanometer

Theory: - MSA (D-6) methanometer is portable, handled, and suitable for spot-checking of gas accumulation with a measurement range from 0 to 5% by volume. The weight of the methanometer is about 0.47 kg. Figure 1 shows MSA (D-6) methanometer, which consists of a Wheat-stone bridge circuit and a very sensitive galvanometer that reads methane percentage directly. Rechargeable batteries are used in the methanometer. The power pack consists of 2 numbers of nickel cadmium batteries in series and provides 2.2 – 2.8 V for operation. Capacity of the power pack is 1.2 Ah (Ampere-hour). Voltage of the battery should be checked before methane check and it must be between 2.2 and 2.8 V. If the voltage is below 2.2 V, the methanometer does not check the methane concentration as the reading is likely to be faulty. The batteries can be charged from a battery charger connected to an A.C. electric supply of 250 V, 50 cycles and the complete charging takes about 14 h. Once the battery is fully charged, it can last for about 500 tests. All the above mentioned components are enclosed in an anti-static plastic case with an opening at the top for sample inlet.

It can detect methane up to 5% with an accuracy of 0.2% for lower concentration (up to 3%).

Specification:-

Weight = 0.47 Kg (with battery)

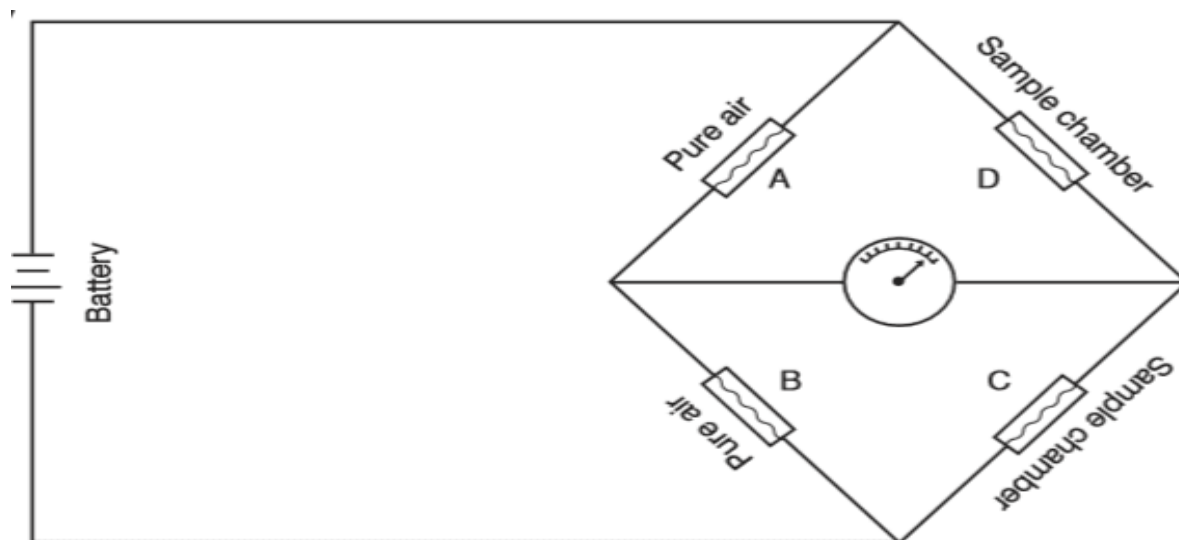
Range = 0-5% of CH₄ in air

Accuracy = 0.2%

Principle of working:

The apparatus utilizes the principle that the resistance of a wire increases on heating by combustion of methane. Thus a balanced Wheatstone bridge circuit becomes unbalanced, passing a current through the galvanometer. The amount of current flow is related to the percentage of methane present in the air.

At the top of this instrument, there is a diffusion head (with dust cover) connected to the aspirator bulb through a rubber tube. The diffusion head is sometimes connected to a telescopic probe (of 1.5 m length) to detect methane (lighter than air) in the roof's cracks. There is a voltage check button pressing at the right end of the instrument, which gives battery voltage 2.2 to 2.8v.



Wheatstone bridge circuit of MSA D6 Methanometer

The apparatus essentially consists of a Wheatstone bridge circuit, and a very sensitive galvanometer graduated to read methane percentage directly.

Air sample is drawn in the detector **(D)** and compensatory **(C)** chamber by an aspirator bulb. The detector filament is made of activated platinum where methane is burnt by the heated filament. Compensatory filament is of de-activated one so that when current

passes through the circuit the temperature of the filament is not sufficient to burn methane. The other two filaments **(A & B)** are housed in sealed chambers containing pure air. Hence, when mine air containing firedamp is drawn, only detector filament resistance changes, making the previously balanced bridge unbalanced. The amount of unbalance indicated by the current flow in the galvanometer is directly calibrated in terms of methane percentage.

Test Procedure:

Step 1: Check the voltage of the battery by pressing the voltage check button. The pointer should lie in the red zone indicating a voltage of 2.2 to 2.8 V.

Step 2: Draw mine air sample by squeezing the aspirator bulb. 2 to 3 squeezes are enough.

Step 3: Press the methane check button for 10- 15 sec. The pointer swings with a jerk and comes back to a steady position momentarily the methane percentage.

From time to time zero error of the scale should be checked in pure air. If necessary it can be adjusted by the zero -setting knob.

Experiment No:-04

Aim: To detect the mine gases, viz. CO, CO₂, O₂, CH₄ and H₂ using Multi-gas Detector.

DRAGER X-AM 7000

Accessories: Drager X-am 7000 Monitor with Internal Pump, integrated rubber boot, carrying strap, standard NiMH battery pack, noted sensors, single-unit battery charger system, calibration adapter, pump adapter with water stop filter

The Drager X-am 7000 Monitor combines electrochemical (EC), catalytic oxidation (CAT), photo ionization (PID) and infrared (IR) sensing technology for a total of up to 5 sensors. There are 25 different electrochemical Drager-Sensors[®], 3 different catalytic sensors, a PID Sensor and 3 different IR sensors.

The Drager X-am 7000 is the smallest gas monitor globally that can utilize all of these technologies in one compact package. The multiple gas measurement and internal pump capabilities make this unit ideal for confined space entry applications like underground mines.

The Drager X-am 7000 Monitor has many “strengths”. The strong internal pump can draw gases from well over 150 ft. (45 m). It is useful for longer boreholes.

The integrated rubber boot and strong advanced polymer enclosure of the Drager X-am 7000 withstand the toughest industrial environments, providing good resistance against dust, rain, and extreme temperatures. Loud audible, and bright visual alarms warn the user that hazardous gas levels are present. The standard NiMH battery pack will last well in excess of an 8-hour shift, and the optional high-capacity battery will allow operation times greater than 20 hours.

Another strength of the Drager X-am 7000 is the optional IR Sensor. The infrared combustible gas (IR-Ex) sensor measures a wide range of hydrocarbons. However, unlike catalytic sensors, the IR sensor will measure hydrocarbons in inert backgrounds (without oxygen). The IR-Ex is very good at measuring heavier hydrocarbons (like turpentine, kerosene and jet fuels) from ppm to %LEL levels. Gases that poison catalytic sensors such as H₂S, halogenated hydrocarbons, and silicones do not affect the IR-Ex Sensor making it ideal for industries that make or use these types of compounds. Gases like methane and propane can be measured in ppm, %LEL, and %Vol. Measurement ranges (with the same instrument) make the X-am 7000 well suited for the natural gas industry.

The IR-CO₂ sensor makes the X-am 7000 the ideal choice for measuring Carbon Dioxide in underground mines. The IR-CO₂ sensor detects from low ppm levels to %Vol. Concentrations of CO₂.

The smart PID sensor measuring a wide range of VOC's up to 5,000 ppm, depending on the gas selected. The Dräger X-am 7000 has 20 compounds pre-programmed in its library and 3 user-programmable entries available. This sensor is well suited for confined space entry, area monitoring and HazMat applications.



TECHNICAL DATA

Smart PID Sensors	VOC's
Smart EC Sensors	O ₂ , H ₂ S, CO, CL ₂ , NH ₃ , NO ₂ , SO ₂ , HCN, CO ₂ , NO, H ₂ , H ₂ S-HC, CO-HC, PH ₃ -HC, H ₂ -HC, ClO ₂ , Organic Vapors, Hydrides, Amines, Mercaptans,
Smart IR Sensors	Combustible Gases (Ex) or Carbon Dioxide (CO ₂)
Smart CAT Sensors	Standard 100 %LEL Combustible Gas or 100% Vol. CH ₄
Size	5.9 x 5.5 x 2.9 inch (150 x 140 x 75 mm)
Weight	2.4 lbs. (1.09 kg) with standard battery
Approvals	UL: Class I, Division 1, Groups A-D, T4 CSA: Class I, Division 1, Groups A-D, T4 ATEX: II2G EEx ia d IIC T4 M2 ia d I

Examples of Volatile Organic Compounds (**VOCs**) are: benzene, ethylene glycol, formaldehyde, methylene chloride, tetrachloroethylene, toluene, xylene, and 1, 3-butadiene.

The gas detector in Dräger X-am 7000 is presented in Figure -1



Figure-1

DrägerSensors

Space for up to 5 sensors, 3 electro-chemical and 2 catalytic bead or infrared sensors or a photo ionization detector (PID).

Alarm functions

360° visible and > 100 dB multi-tone audible alarm.

Robust enclosure

Rugged, water-tight with a standard integrated rubber boot.

Large display

Clearly structured, scratch resistant display with information in plain text.



Infrared sensors: The principle of IR sensors lies in the absorbed radiation by hydrocarbons in the wavelength range corresponding to infrared. An IR emitter is installed inside the sensor. It emits waves, which are reflected and directed to the detector. The gaseous mixture passes through the flame arrestor into the sensor. The major air components, i.e. oxygen and nitrogen, do not absorb the radiation on the detector as far as infrared is concerned, and the intensity or radiation on the detector may be weakened only in the presence of hydrocarbons. If so, an electrical signal is emitted, and it is transmitted to the display of the measuring device, e.g., 0–100 vol. % of gas or % of the lower explosive limit. The principle of measurement of flammable gas concentration in an infrared sensor is presented in [Figure 2](#).

IR sensor

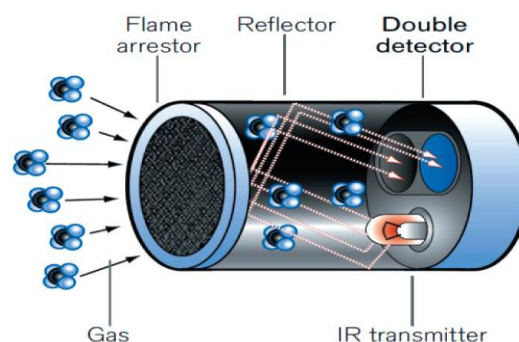
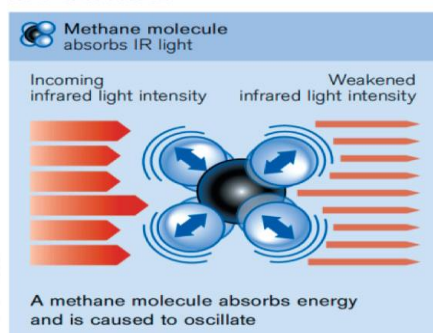


Fig. 2. Principle of infrared sensors in Multi-gas Detector

The infrared sensors IR Ex in the gas detector are used for measuring the concentration of carbon dioxide and explosive gases, e.g. methane, propane, and ethylene. The IR Ex sensor allows for measuring methane concentration both within the lower and upper explosive limit and in the range of 0 to 100 vol. % with the accuracy of 0.01 vol. %. The IR CO₂ for measuring the carbon dioxide concentration can be used for measurements of 0 to 5 vol. % of CO₂ in the air.

Electrochemical sensors are microreactors generating a small current in the presence of reactive gases. It consists of at least two electrodes – a measuring electrode and counter electrode. They are connected in two ways: on one hand by the electrolyte (fluid conducting ions), and on the other one, by an external electrical circuit. Gas inflows through a porous membrane to the sensor filled with the electrolyte, and current is generated between the electrodes. The intensity of measured current is transmitted to the gas concentration display in the detector. The principle of electrochemical sensors installed in Dräger gas detector is presented in Figure 3.

Electrochemical sensor

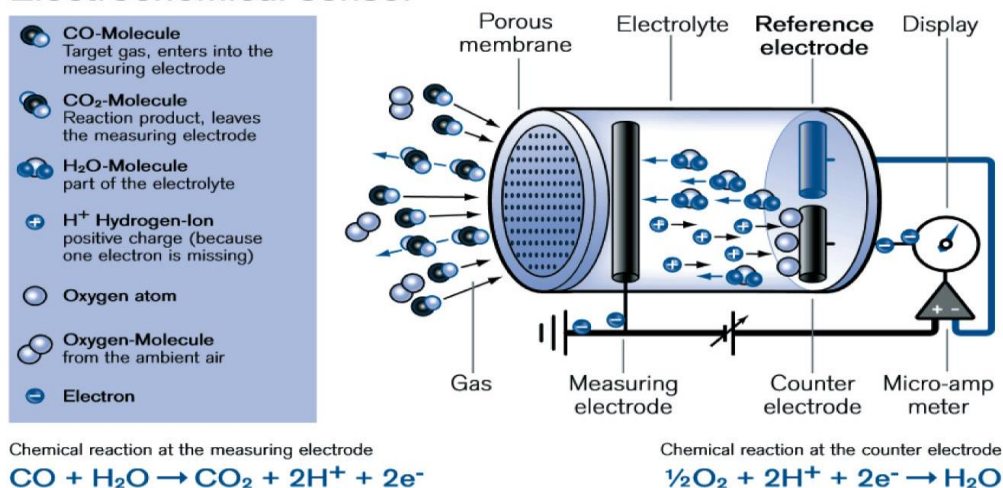


Fig.3. Principle of electrochemical sensors in Multi-gas detector.

Range of Sensors:-

Infrared Sensors

CO₂ -- 100 Vol%
CH₄ -- 100 Vol%

Electrochemical Sensors

O₂ -- 25 Vol%
CO -- 9900 ppm
H₂ -- 1000 ppm

Experiment No:-05

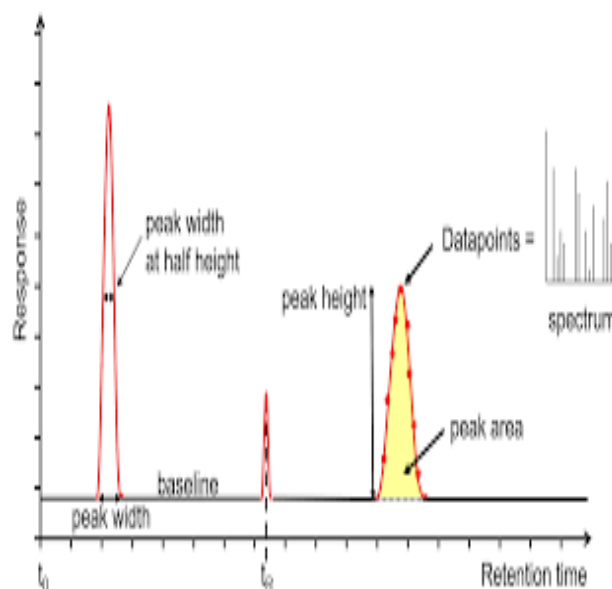
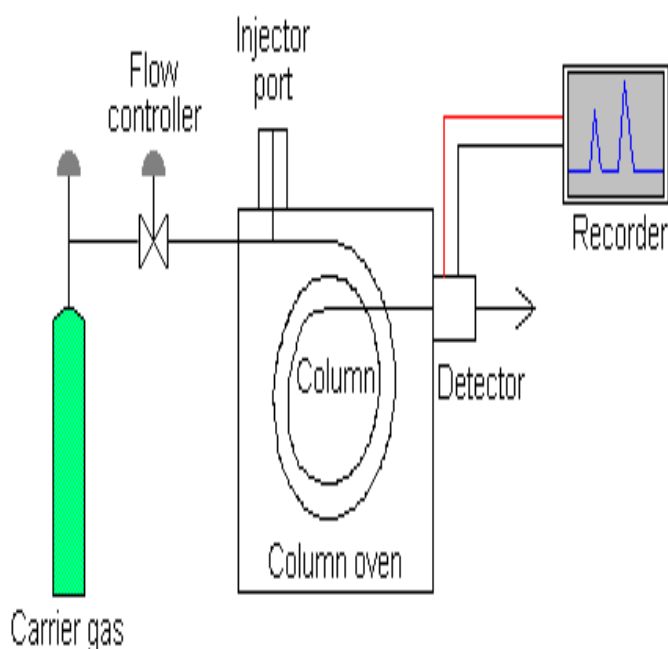
Aim: - Analysis of mine air using Gas Chromatograph

Theory: - Chromatography is a physical separation method in which the separated components are distributed between two phases, one of which is stationary while the other moves in a definite direction.

In gas chromatography, the mobile phase (or moving phase) is a carrier gas, usually an inert gas such as helium or an unreactive gas such as nitrogen. The stationary phase is a microscopic layer of liquid or polymer on inert solid support inside a piece of glass or metal tubing called a column. The instrument used to perform gas chromatography is called a gas chromatograph.

The basic parts of the GC:

1. Pneumatic system
2. Injection system or the sampling system
3. The Columns
4. The Detector
5. The recorder / Data Processor system



Principle: - It involves passing a mixture dissolved in a mobile phase through a stationary phase which separates the analyte to be measured in the mixture based on differential partitioning between the mobile & stationary phase.

The gaseous compounds being analyzed interact with the walls of the column, which is coated with a stationary phase. This causes each compound to elute at a different time, known as the compound's retention time. The comparison of retention times is what gives GC its analytical usefulness.

Terms:-

- **Chromatograph:** - An equipment that enables sophisticated separation of gas or liquid.
- **Analyte:** - The substance that is to be separated during chromatography.
- **Bonded phase:** - A stationary phase that is covalent bonded to the inside wall of the column tubing.
- **Chromatogram:** - The visual output of the chromatograph is known as chromatogram. Different peaks or patterns on the chromatographs correspond to the separated mixture's various components.
- **Retention Time:** - The characteristic time taken by a particular analyte to pass from the column inlet to the detector.
- **Mobile Phase:** - The phase which moves in a definite direction may be liquid or gas. It consists of the sample being separated, analyzed & the solvent that moves the sample through the chromatography column.
- **Stationary Phase:** - The substance is fixed in place of the column tubing.
- **Sample:** - The matter is analyzed in chromatography. It may consist of a single component or a mixture of compound. Generally, a chromatograph is composed of carrier gas supply, sample inlet & Column
- **Carrier gas supply:** - A mobile phase, an inert gas, is delivered by a gas generator or a gas cylinders. The most common carrier gases are N₂, H₂ & He. These inert gases should have high purity and thermal conductivity.

- **Sample injection port:** - The most common injection method is where a micro syringe is used to inject a sample through a rubber septum in a flask vaporizer part the heads of the column. This injector can be used in one of the two modes split or splitless. This injector contains a heated chamber containing a glass liner in which sample is injected through the septum.
- **Column:** - The column of a GC acts as the heart of the equipment. It is responsible for the separation of individual constituents of a gaseous mixture. It contains and adsorbent which retains different gases of different duration in the column. These are of 2 types: Packed & Capillary.
- **Detector:** - The most commonly used detectors are the flame ionization detector (FID) and the thermal conductivity detector (TCD). Both are sensitive to a wide range of components, and both work over a wide range of concentrations.
- **Recorder:** - The recorder serves the function of identification and quantification of various components of the mixture to be analysed.



Experiment No:-06

Aim: - Measure for airflow rate in U/G mine opening using vane Anemometer & Velometer.



Anemometer:-

It is an apparatus used to determine the air velocity directly or indirectly in a stipulated time. It consists of a 100 mm diameter cylindrical structure enclosing a hub on which eight-light metal vanes are mounted at an angle of 45° to the airflow direction. The vane circle operating through gearing rotates at speed proportional to the air velocity. A counter located at the center of the hub indicates the number of revolutions of the wheel, or distance travelled by the vanes in meters on dials. Clutch with a lever is provided to allow starting and stopping of the counting mechanism at will so that the instrument reading for one or two minutes as noted with a stopwatch, can be obtained

from which air velocity in m s^{-1} can be calculated by dividing the reading with the duration of the measurement. However, in some cases, an in-built stopwatch mechanism is fitted, which automatically engages the counting mechanism for one minute, and hence air velocity can be obtained directly. When the clutch is engaged, the inertia of the counting gear act as a brake on the vanes and a small time is required before the vanes again indicate the correct speed. The slow-motion period becomes essential if the anemometer is being used for a short period of observation, say less than 20 seconds.

The bearing friction of the vane wheel and the counting gears also cause the vanes not to rotate below a specific velocity range depending on the type of bearings used.

These anemometers commonly used in mines are suitable for measuring air velocities from 1 to 15 ms^{-1} . Low-speed models having jeweled bearings are suitable for the velocity range of 0.15 to 1.5 m s^{-1} & high-speed models are suitable for 10 to 30 ms^{-1} velocity.

Measurement: - For measurement with an anemometer, three advisable methods are there:

- (a) **Single Point Method**: - This method does n't give the average velocity. It provides instantaneous velocity. To get average velocity, we have to multiply by a factor 0.8 to 0.9, only when the instrument is kept at 1/7 of Dia of opening, thus giving method Factor=1.
- (b) **Continuous Traversing**: - Here anemometer is traversed slowly, usually from top to bottom and then to top, over the whole cross-section of the airway during the measurement to get an average velocity at the place. This is used where the desired accuracy is most high.
- (c) **Precise Traversing**: - In this method, the cross-sectional area is divided into small areas. The observations at each small area in the airway are taken to obtain an average from these observations.

Procedure

- (I) First, the cross-section area of the airflow tube is measured.
- (II) Then continuous Traversing method is followed.
- (III) The vane anemometer is placed perpendicular to the airflow along the face of airflow in a particular pattern.
- (IV) Stopwatch is started, and the vane anemometer is moved along the face of airflow in a particular pattern.
- (V) The reading is recorded
- (VI) The process is repeated for four times, and the average velocity is taken.

Calculation: - Let Cross section area (**A**) = 4 m², Velocity (**V**) = 110 m/min.

Then Quantity of air flow = $A \times V = 4 \times 110 = 440 \text{ m}^3/\text{min}$.

Work Place Section: - The place where the average velocity of air has to be measured should be selected in such a way that

- a. The roadway should have a nearly uniform cross-section for almost 15m on either side and should be straight.
- b. The cross-section should be such that its area can be calculated.
- c. It should be away from bends and junctions and free from obstruction, which may cause turbulent airflow.

Precaution:-

- a. The presence of an operator in the roadway gives rise to the obstruction of the airflow and distorts the velocity pattern in the airway. Hence anemometer rod should be used to suspend it freely.
- b. The plane of traverse of the anemometer should be totally at the right angle to the anise of the roadway.
- c. Traversing should be done at a slow speed to eliminate errors in the estimation of air velocity.
- d. Anemometer calibration should be carried out before every ventilation survey.

Velometer: - The Velometer is a swinging vane instrument in which the air velocity is obtained from the degree of deflection of a spring balanced aluminum vane mounted in jeweled bearings in a rectangular case. The vane moves in a rectangular passage. The deflection of the vane is proportional to the air velocity. The deflection is read from the

movement of a connected pointer on a graduated scale fixed on the top of the instrument. The use of delicate bearings and springs makes the velometers extremely sensitive, and generally, a minimum reading of 0.15 m/s is possible with the instrument. The provision of inter-changeable jets for fitting into the inlet port permits an increase of the upper measurement limit to 30 m/s.

These are direct reading instruments where no calculations or calibration chart is necessary. It is a spot-reading instrument and can't be used for continuous traversing.

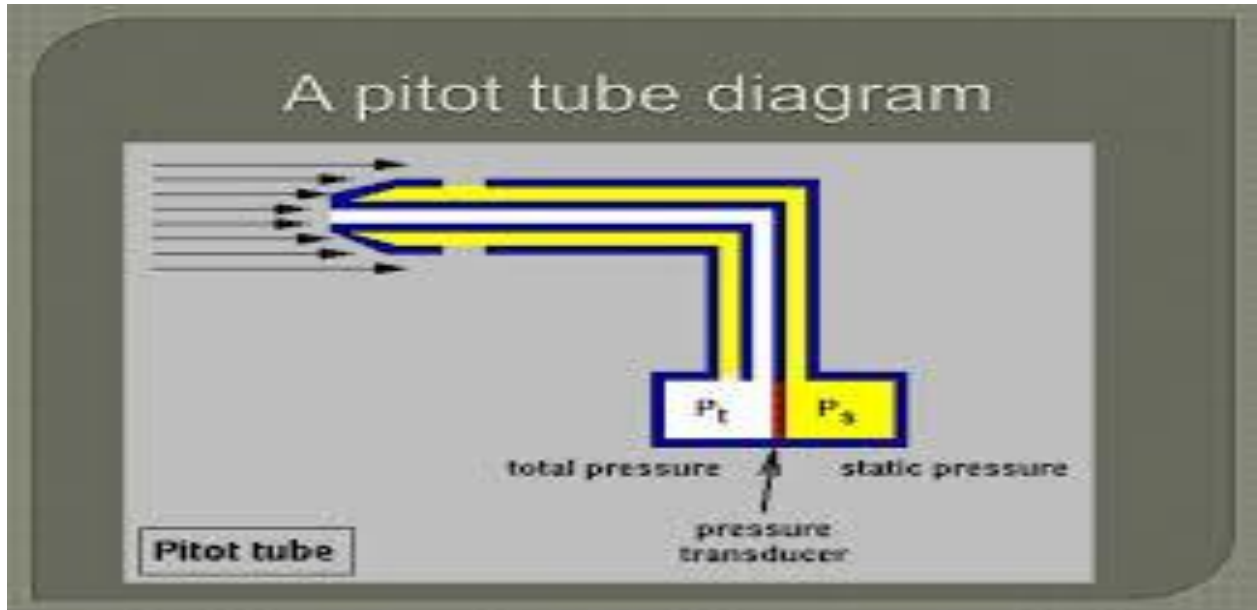


Velometer

Experiment No:-07

Aim: - Measurement of air gas flow & air pressure using Pitot's tube.

Apparatus: - Pitot static tube.



Pitot tube: - A single-point airflow measuring instrument is used to measure pressure. It can measure air velocity indirectly. It is bent at a right angle and consists of internal and external tubes. Several holes at the periphery of the outer tube, generally known as a static tube, measure static pressure. The inner tube measures total pressure.

Theory: The pitot-static tube comprises external and internal tubes. The two tubes are connected to the pressure gauge to measure pressure. It measures static pressure from the static tube, total pressure from the inner tube, and the difference of total pressure and static pressure gives velocity pressure.

$$P_V = P_T - P_S$$

Where, P_V = Velocity pressure of air

P_T = Total pressure of air

P_S = Static Pressure of air

The velocity can be obtained from the velocity pressure by using the relation.

$$P_V = \frac{KV^2\rho}{2}$$

Where, P_V = velocity pressure of air in Pa

ρ = Density of air = 1.2 kg/m^3

K = correction factor for the particular pitot

Static tube (for standard designs, $K = 1$)

After putting standard values

$$V = 1.29 \times \sqrt{P_v}$$

The velocity obtained by this may not be very accurate because some head losses are there due to the bending of the tube. Also, the yaw angle should be as small as possible.

Procedure:-

- (a) The head of the Pitot tube is kept parallel to the airflow direction keeping its head opening along the airflow direction.
- (b) Connect the other ends of the static and internal tube to the pressure gauge and note down the reading for different sized resistances in the air flow direction.
- (c) Calculate the velocity pressure and velocity using the formula.
- (d) Next, the static tube is connected to the same and the velocity of airflow is measured using the formulae.

Precaution:-

- (a) The angle of yaw must be kept within $\pm 10^\circ$
- (b) The reading should be taken only through the low-pressure tube of the pressure gauge.

Experiment No:-08

Aim: - Determination of psychrometric properties of air.

Theory:

Psychrometry is the physical and thermodynamic properties of dry air and water vapour mixtures. Certain psychrometric terms are used to describe the moisture content in the atmosphere, i.e., a dry air-water vapour mixture. Psychrometry is made up of two words “Psychro,” which means cold and “metron” which means measurement.

Psychrometric properties are measured from:

1. Barometric Pressure
2. Dry Bulb Temperature
3. Wet Bulb Temperature

Calculation

Saturated vapour pressure (S_{vp}): This is the maximum pressure that can be exerted by the water vapour in a saturated atmosphere corresponding to a particular temperature. S_{vp} in terms of kPa can be calculated using the following formula:

$$S_{vp} = 0.6105 \times \exp\left(\frac{17.27 \times t}{237.3 + t}\right)$$

where S_{vp} is the saturated vapour pressure (kPa) and t is the temperature ($^{\circ}\text{C}$).

Vapour pressure (V_p): This is the partial pressure of water vapour present in a certain volume of air. This is also sometimes called as actual vapour pressure. V_p can be approximately determined from the following formula:

$$V_p = (S_{vp})_{wb} - 0.000644 \times P \times (t_{db} - t_{wb})$$

and

$$(S_{vp})_{wb} = 0.6105 \times \exp\left(\frac{17.27 \times t_{wb}}{237.3 + t_{wb}}\right)$$

where V_p is the actual vapour pressure (kPa), $(S_{vp})_{wb}$ is the saturated vapour pressure at wet bulb temperature (kPa), P is the barometric pressure (kPa), t_{db} is the dry bulb temperature ($^{\circ}\text{C}$) and t_{wb} is the wet bulb temperature ($^{\circ}\text{C}$).

Relative humidity (RH): RH is the ratio of actual vapour pressure to the saturated vapour pressure at the given dry bulb temperature. It can be calculated in terms of percentage using the following formula:

$$RH = \left(\frac{V_p}{(S_{vp})_{db}} \right) \times 100$$

and

$$(S_{vp})_{db} = 0.6105 \times \exp \left(\frac{17.27 \times t_{db}}{237.3 + t_{db}} \right)$$

Mixing ratio (M): This is the ratio of mass of water vapour present to the unit mass of dry air. Mathematically,

$$M = 0.622 \times \frac{V_p}{P - V_p}$$

This is often expressed as moisture content in kg of moisture per kg of dry air.

Specific humidity (SH): This is the mass of water vapour present in kg per kg of moist air.

$$SH = \frac{0.622V_p}{P - 0.378V_p}$$

Dew point (DP): The temperature at which the air attains saturation is known as its dew point. It can be calculated using the following formula:

$$DP = \frac{237.3X}{17.27 - X}$$

where

$$X = \ln \left[\frac{V_p}{0.6105} \right]$$

Density of unsaturated air (kg/m³): This can be calculated from the barometric pressure, temperature and vapour pressure of the unsaturated air from the following formula:

$$\rho = \frac{P - 0.378V_p}{0.28704 \times (273.15 + t)}$$

Apparent density (kg of dry air/m³ of air-water vapour mixture): This can be calculated:

$$\rho = \frac{P - V_p}{0.28704 \times (273.15 + t)}$$

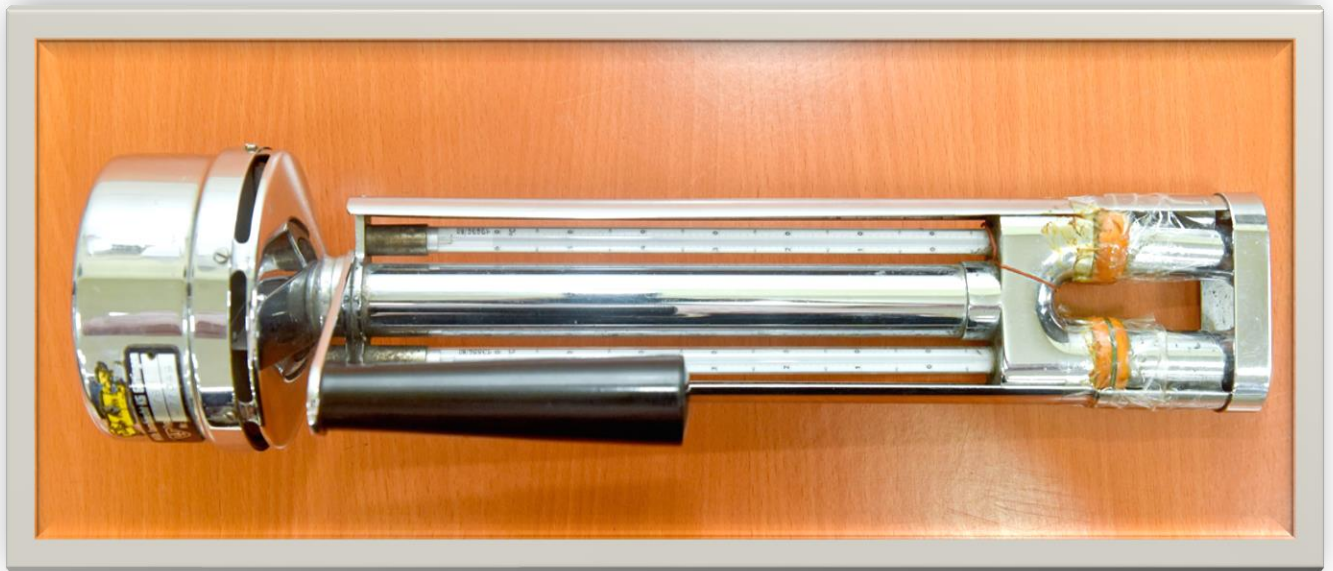
Specific volume (m³ of dry air /kg of dry air): it is reciprocal of density.

$$V_m = \frac{0.28704 \times (273.15 + t)}{P - 0.378V_p}$$

Apparent specific volume (m³ of air-water vapour mixture /kg of dry air): It is reciprocal of apparent density. It is defined as the total volume of dry air and water vapour mixture per kg of dry air.

$$V_m = \frac{0.28704 \times (273.15 + t)}{P - V_p}$$

Measurement of Humidity using Assmann Psychrometer/Whirling Hygrometer:-



Assmann Psychrometer



Whirling Hygrometer

Whirling hygrometer: Whirling hygrometer is commonly used in mines for better accuracy. It consists of two thermometers mounted side by side, the bulb of one being exposed to the air and that of the other covered with a thin gauze of muslin, the lower end of which is dipped in a container filled with water. Distilled water should be used in the container as impurities in water affect its rate of evaporation. The surface of the muslin should be clean, and the muslin should be changed from time to time.

Precautions should be taken to take the thermometer reading as quickly as possible after stopping the rotation of the hygrometer. As the wet-bulb temperature gets affected most by stagnant air, the wet-bulb reading should be taken first.

Assmann psychrometer: This is an aspiration psychrometer with the bulbs of the thermometer ventilated by a fan operated by a clock-type spring. The bulbs of the thermometer are covered with metal sleeves whose outer surface are bright chromium-plated in order to prevent radiation from surrounding surfaces affecting the readings of the thermometers. For the same reason, it is desirable to hold the psychrometer away from the body of the observer and other surfaces. The fan should be run for three minutes before taking a reading. For accuracy, an occasional checking and control of ventilation speed is essential.

Observation:-

Wet bulb temperature of air in room = 23.5°C

Dry bulb temperature of air in room = 28.3°C

Result:-

The results are as follows:

Vapour pressure = 2.58 kPa, Relative humidity = 67%, moisture content = 0.0163 kg/kg, specific volume = $1.13\text{ m}^3/\text{kg}$, Enthalpy = 70.13 kJ/kg

Precaution:-

1. The bulb of the psychrometric should not be touched as it may change the temperature.
2. The psychrometer should be held away from the body and other surfaces

Experiment No:-09

Aim: - Measurement of air cooling power using Kata Thermometer.

Apparatus: - Kata thermometer

- Stopwatch
- Hot water in thermo flask
- Wet muslin Cloth.



Kata thermometer



KATA THERMOMETER

KATA THERMOMETER MEASURES THE COOLING POWER OF THE ENVIRONMENT; IT IS USED TO ESTIMATE THE PERSONAL COMFORT OF WORKERS (SEE ALSO "HEAT STRESS MONITOR" AND "PERSONAL TEMPERATURE MONITOR"). A SPIRIT-IN-GLASS THERMOMETER IS USUALLY USED: ITS BULB IS HEATED TO ABOVE BODY TEMPERATURE, REMOVED FROM THE HEAT SOURCE AND ALLOWED TO COOL. THE TIME TAKEN FOR THE THERMOMETER READING TO DROP FROM ABOVE TO BELOW NORMAL BODY TEMPERATURE (E.G., FROM 38 °C TO 35 °C) IS USED TO CALCULATE THE COOLING POWER OF THE ATMOSPHERE, USING A CALIBRATION FACTOR F ENGRAVED ON THE THERMOMETER BY THE MANUFACTURER:
COOLING POWER = F/(TIME OF COOLING).

Kata Thermometer

- ✓ A spirit-in-glass thermometer with a large bulb.
- ✓ Two marks on the stem at 35°C and 38°C.
- ✓ The thermometer is warmed to over 40°C, placed in the wind and the time taken for the spirit column to fall from 38°C to 35°C is measured. The wind speed can be derived from: $B^2V = \left(\frac{F}{(36.5 - T) \cdot t - A} \right)$ where F is °F and t is in seconds. A and B are constants determined from calibration.

Construction & Theory: In 1916, Professor Leonard Hill introduced the Kata thermometer to determine the comfort condition of an environment that considers all the heat stress parameters. Kata thermometer is a method used to express the comfort condition of an environment that considers all the heat stress parameters. It consists of a glass thermometer with a

cylindrical bulb of 2cm diameter and 4cm length filled with coloured alcohol connected to a 20 cm long stem having two smaller bulbs or reservoirs at the top and bottom ends. The stem has two marks at 35 °C (308 K) and 37.8 °C (311 K) in between the two bulbs. While taking a reading with the Kata thermometer, the bulb is first heated by dipping it in hot water so that the liquid rises to the top reservoir. The bulb is then taken out of the water, wiped with a dry cloth and is allowed to cool in the mine air. The time taken by the thermometer to cool from 37.8 °C to 35 °C is noted. The Kata cooling factor, which gives the amount of heat lost per cm² of the surface of the bulb in cooling from 37.8 °C to 35 °C, is usually written on the Kata thermometer. This factor divided by the time taken for the cooling gives the rate of heat loss from the Kata thermometer.

The dry-kata reading estimates the heat loss from the surface of the larger bulb due to radiation and convection. It hence is of little importance, particularly under hot and humid conditions where most of the heat loss from the human body is through evaporation.

The wet-kata reading is obtained by covering the bulb of the Kata thermometer with wet muslin. The idea is to resemble the human body, which loses heat by radiation, convection, and evaporation. The wet-kata cooling power is related to the wet-bulb temperature and the air velocity by the following relation:

$$K = (14.65 + 35.59v^{1/3}) (309.65 - T_w), \text{ for air velocity } v \leq 1 \text{ m/s}$$

$$K = (4.19 + 46.05v^{1/3}) (309.65 - T_w), \text{ for air velocity } v \geq 1 \text{ m/s}$$

Where,

K = Kata cooling power (W/m²)

v = velocity of mine air (m/s)

T_w = wet bulb temperature (K)

Wet-Kata reading can be approximated from air velocity v and wet-bulb temperature T_w

$$\text{wet kata cooling power, } K = (0.7 + v^{0.5}) (36.5 - T_w)$$

Where,

K = wet kata cooling power (mcal/ (cm²s)),

T_w = wet-bulb temperature (°C)

v = velocity of air current (m/s)

Procedure:-

- The setup is arranged, and two readings are taken; one with outer air current and the other with air current.
- The time taken for the alcohol column to drop from 38 °C to 35 °C mark was noted down.
- For measuring the wet kata cooling power, a wet-muslin cloth was wrapped around the bulb of the kata thermometer.
- By dividing the kata factor with the time recorded above, air cooling power is obtained.

Calculation:-

Air cooling power (W/m²) = kata factor/time taken

Kata factor = 307 milli-calories /cm²

Time taken by dry kata thermometer = 156 seconds

Time taken by wet kata thermometer = 77 seconds

Air cooling power by dry kata thermometer = 307/156

$$\text{Thus} \quad \frac{307 \times 4.185 \times 10 \times 5}{156} = 411.8 \text{ W/m}^2$$

Air cooling power by wet kata thermometer = 307/77

$$\text{Thus} \quad \frac{307 \times 4.185 \times 10 \times 5}{77} = 834.28 \text{ W/m}^2$$

Result: - Air cooling power by dry kata and wet-kata readings are 411.8 W/m² and 834.28 W/m², respectively.

Limitations:-

- Variation of kata factor with temperature.
- Kata thermometer underestimates the effect of temperature and humidity of mine air and overestimates the impact of air velocity.
- The convective and evaporative coefficients of the kata thermometer are twice that of the human body, and therefore, it cools much faster than the human body.

- Kata thermometer has a smaller volume to surface area ratio compared to the human body, which further makes it cool faster than that of the human body.
- Its use as a heat index is limited to hot and humid climates.
- A person doing moderate work generally produces 165 W/m^2 . A kata reading of 165 W/m^2 will provide a comfortable working environment. However, it is not the case. As discussed earlier, the kata thermometer cools much faster than that of a human body, and therefore, for a comfortable working condition, 165 W/m^2 should be multiplied by a factor. This factor has arrived at 5 through various experimental procedures. Therefore, a kata reading of $5 \times 165 \text{ W/m}^2$ is essential to provide a comfortable working environment.
- The larger bulb (or reservoir) is fragile.

Experiment No:-10

Aim: - Study of fan characteristics curves.

Apparatus required : Fan duct, pitot tube, pressure gauge, Different resistances

Background

The performance of a fan under various conditions can't be expressed by simple mathematical equations but is most conveniently expressed by a series of curves known as characteristic curves. These curves consist of system characteristic curves and fan characteristic curves.

Theory: - A fan characteristic curve is a curve that shows how the magnitude of one quantity varies with changes in some other related quantity. In the case of a fan for every speed, a set of curves can be drawn to show the variation of fan-draft pressure, B.H.P. of the prime-mover, and mechanical efficiency of the fan with changes in the volume of air. These curves indicate the performance or characteristic of the fan and hence, are referred to as the 'characteristic curves' of that particular fan. Sometimes the term "fan characteristic" denotes the first (pressure-volume curve) of the set of these curves.

Centrifugal fan: - On the basis of blade shapes, the centrifugal fans can be subdivided into (a) forward bladed, (b) radial bladed, and (c) backward bladed fans. Generally speaking, it is the shape of the blades which determines the relative performance characteristic of fans. The theoretical head developed in a centrifugal fan can be calculated with the help of velocity vector diagrams at the inlet and outlet tips of the fan blade.

Axial flow fan: - An axial flow fan consists of one or more rotors equipped with a set of blades mounted on a rotating shaft and enclosed by a casing. During the passage of air through the impeller, the curvature and rotation of the blades cause a reduction in the relative velocity of air with respect to the blade. Axial flow fan blades are usually of aerofoil section, although curved plates behaving like an aerofoil can also be used with reduced operational efficiency.

Fans in series: - In a mine of high resistance, a series combination of mine fans increases air quantity, which is possibly more than 30%. When two fans are added in series, a combined pressure-volume characteristic is obtained.

Fans in parallel: - In mine, with a large quantity requirement at relatively low pressure, parallel operation of two or more fans can be considered. The combined characteristics curve is obtained by jointed (added) the quantity value for each fan for the same common pressure.

Comparison of Fan in series and parallels combination:-

- In a mine of high resistance, the series combination gives a considerable increase in quantity of air flowing (possibly more than 30% increase) while parallel combination gives negligible increase.
- In a mine of medium resistance both the series and parallel arrangement gives the same result in the region of about 20% more quantity.
- In a mine of low resistance where a single fan is operating at a very low efficiently on an arrangement, there the series will give low quantity and parallel arrangement is preferred.

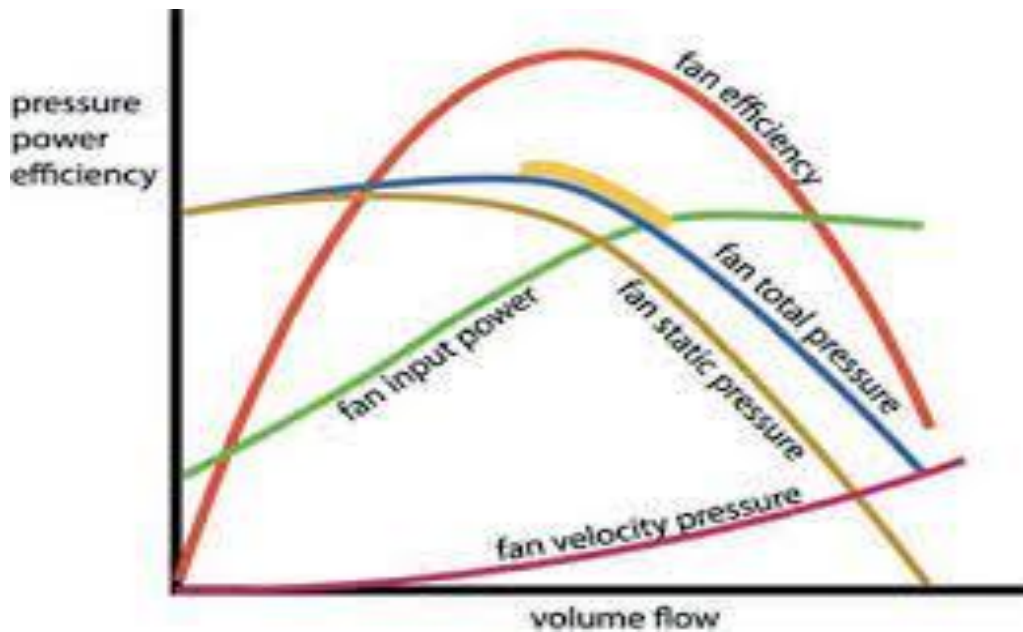
Pressure curve: - variation in pressure developed by a fan with changes in the volume of air

Input power curve: - The power required at the fan shaft with varying air quantity of air delivered.

Efficiency Curve: - The efficiency curve of a machine or equipment is defined as the ratio of useful work to the energy input. The efficiency of curve is basically derived from power input curve and the static curve.

Air Power = Pressure X Quantity

Fan efficiency = Air Power/ Input power of fan



Full set of fan characteristic curves

Procedure:

Procedures

- i. First, pitot tube is placed in the fan duct for measuring the fan pressure.
- ii. During measurement, it must be kept at the lowest angle of yaw between the direction of airflow and the horizontal axis of head of the pitot-tube.
- iii. For measuring the total pressure, the inner tube is connected to one limb of the pressure gauge while the other limb is open to the atmosphere.
- iv. For measuring static pressure, only outer tube is connected to the pressure gauge.
- v. For measuring velocity pressure, both inner tube and outer tube are connected to both legs of the pressure gauge. Or the velocity pressure can also be obtained by subtracting the static pressure from total pressure.
- vi. The velocity of air can be calculated by using the following relation:

$$v = \sqrt{\frac{2P_v}{K\rho}}$$

Where v is the velocity of air (m/s), P_v is the velocity pressure (Pa), ρ is the density of air (kg/m^3) and K is the correction factor for the particular pitot-static tube (for standard design, $K = 1$)

- vii. Mechanical efficiency of fan can be calculated by dividing the air power with the power input to fan motor.
- viii. The above points (i.e. iii, iv, v, vi and vii) will be repeated at different air flow rates, which can be obtained by the multiplying the velocities of air and cross-sectional area of fan duct. The airflow rates will be varied by providing the different resistances at the mouth of the fan duct.
- ix. Fan pressure, the brake horse power of the drive motor and mechanical efficiency of the fan will be plotted as a function of airflow rate on the graph paper.



Series and parallel fan